KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

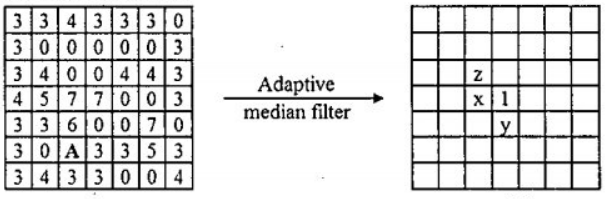
(Answer **ANY THREE** questions from this section in Script A)

1. a) Compare and contrast between image processing and computer vision. Mention some applications of them. (08)
   b) Use Laplacian operator to sharp the following image. Use necessary border reflection padding. (15)

$$\begin{bmatrix} 1 & 2 & 5 & 4 \\ 5 & 2 & 5 & 5 \\ 2 & 1 & 6 & 3 \\ 2 & 4 & 7 & 6 \end{bmatrix}$$

   c) What is periodic noise? Design a band reject and a notch reject filter to remove such noise from image. Also discuss their pros and cons. (12)

2. a) Given an input image of size 7x7 shown below, was filtered using 3x3 adaptive median filter with maximum allowed size of 5x5. (12)



input image output image

Answer the following questions with reasoning:
i. What is the value of the pixel A in the input image?
ii. What are the values of pixels x, y and z in the output image?

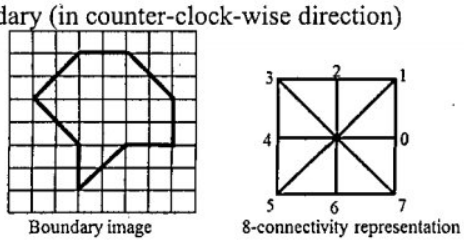
   b) Explain why and how non-maximum suppression is performed in Canny edge detection algorithm. (10)
   c) Let a 3-bit grayscale image is given below: (13)

0	5	2
6	1	7
3	4	7

Resize the image by using bilinear interpolation and backward mapping to obtain 3x5 image.

3. a) Show that $\nabla^2 G = \frac{1}{\pi\sigma^4}\left[1-\frac{x^2+y^2}{2\sigma^2}\right].e^{-\frac{x^2+y^2}{2\sigma^2}}$, where G is the Gaussian function and ∇^2 is the Laplacian operation. (08)
   b) What is Thresholding? List the steps of Otsu's Thresholding algorithm. (11)
   c) Compare the first order and second order derivative based edge detection methods. (10)
   d) State and prove the separability property of 2D DFT. (06)

4. a) For the following boundary (in counter-clock-wise direction) (09)



Boundary image 8-connectivity representation

compute the 8-connected
i. chain code.
ii. chain code that is invariant to rotation.
iii. chain code that is invariant to both starting point and rotation.

b) Consider two consecutive frames of a video shown below. (14)

y \ x	0	1	2	3	4
0	3	15	6	4	0
1	4	4	12	1	2
2	4	14	5	5	8
3	6	1	13	2	3
4	10	7	11	9	12

Frame - 1

y \ x	0	1	2	3	4
0	4	6	5	3	0
1	2	3	15	6	4
2	3	4	5	12	1
3	10	4	13	5	5
4	9	6	1	13	2

Frame-2

Estimate the optical flow of the center pixel using Lucas-Kanade method. Use 3x3 window for spatial coherence.

c) Explain inverse filtering process to restore a noisy image. What is the shortcoming of inverse filter? How does Wiener filter overcome such shortcoming? (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is digital image and digital image processing? What are the fundamental steps in digital image processing? (10)

b) Define blind spot and brightness adaption. Briefly describe how to characterize brightness discrimination using appropriate example. (08)

c) What do you mean by image compression? Briefly describe about Discrete Cosine Transformation of image compression. (12)

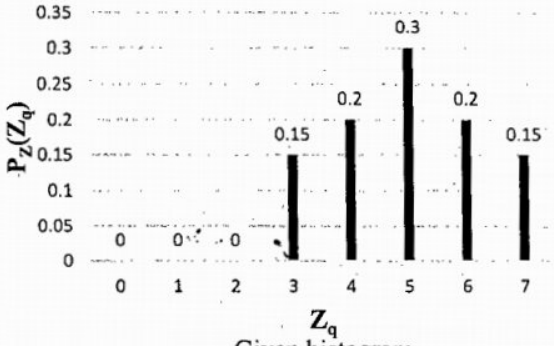
d) What are the changes occur in an image by changing values in spatial domain and quantization (Gray value)? (05)

6. a) Define contrast. Design a mapping function to enhance the contrast of an image having very bright pixels. (11)

b) Briefly describe Power-law transformation for digital image processing and also show the image appearance of different values of gamma. (08)

c) The following table shows the information of a 3-bit 64x64 image. (16)

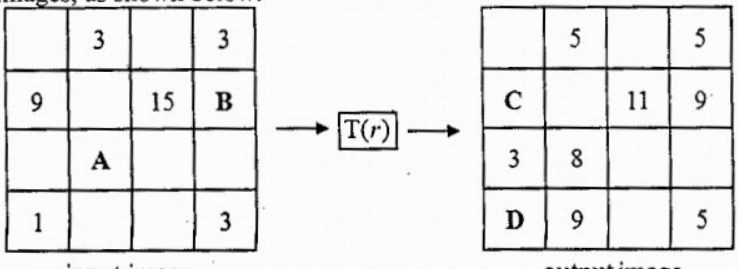
Pixel Value	Frequency
0	790
1	1023
2	850
3	656
4	329
5	245
6	122
7	81



Given histogram

Draw the original histogram, normalized histogram and finally show the matching histogram with the given histogram.

7. a) Let a 4-bit, 4x4 image passes through an intensity transformation function given by $s = T(r) = \alpha \log_2(1+r) + \beta$, where α, β are constants. Few pixels are available in the input and output images, as shown below. (10)



input image output image

What are the values of the pixels A, B and C, D in the input and output images respectively?

b) Define Pseudo color image processing. Discuss about intensity slicing with appropriate diagram. (10)

c) Write down the procedure to compute pruning of a binary image. (05)

d) Consider the following 5x5 8-bit image with grayscale range of [0, 9]. (10)

$$\begin{bmatrix} 0 & 5 & 7 & 7 & 5 \\ 7 & 2 & 6 & 2 & 6 \\ 6 & 9 & 7 & 7 & 0 \\ 6 & 6 & 1 & 7 & 6 \\ 9 & 3 & 2 & 4 & 8 \end{bmatrix}$$

Compress the image using Huffman encoding techniques and also discuss the efficiency of this compression.

8. a) Consider the following binary image A. (12)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Image A

Show the result of applying the following operations on image A with a 3x3 structuring elements.

i. $B = (A \oplus S) \cap A^C$

ii. $B = (A \ominus S) \cup A^C$

iii. $B = (A \oplus S) \cup A^C$

b) How can you detect how many template images are there in a full image using hit or miss transformation algorithm? Briefly describe with appropriate example. (11)

c) Define Hue and Saturation. Write down the process of conversion of a pixel value from HSI color space to RGB color space. (12)